

DATA 641

Word meaning

Before we begin

- Assignment 1 is due today!
- Last lecture, we talked about what constitutes a word, MWE/collocations, and statistical hypothesis testing

Word Meaning

- **Word** - See prior lecture for the hidden complexity!
- **Meaning** - What do we mean by meaning?

Lexicography

The Lexicographic Tradition

- The **lexicographic tradition** involves constructing dictionaries
- Dictionaries enumerate word meanings (senses)
- What are definitions?
 - Definitions are not “defining” in the mathematical sense
 - And also not in a precise legal sense
 - Example: legal definition of “assault” - intentional, affects another person with reasonable apprehension, harmful or offensive contact, imminently, physical injury NOT required
- “Definitions” are text you read that evoke concepts in your head

Forms of word sense distinction

- Homonymy
 - “Pen”
 - “Place to keep animals”
 - “Instrument to write with”
 - Coincidental or historical?
- Polysemy
 - “Bass”
 - “Low register”
 - “Person who sings low”
 - Related meanings
- Systematic / Structured polysemy
 - A form of polysemy that applies systematically to a category
 - Example: *chicken*, *lamb*, *goose*, etc. all have a sense that corresponds to the animal itself, and another that corresponds to the food made from that animal.

Lexicography - making dictionaries

- Labor-intensive tradition
- Ex. Oxford English Dictionary (O.E.D.), started 1857
 - Crowdsourcing by volunteers to find quotations illustrating each word took more than 50 years to complete!
 - cf. Simon Winchester, *The Professor and the Madman*
- Corpus analysis approach
 - Ex. Collins COBUILD dictionary, published 1987
 - KWIC: Keyword in context
 - ... sing a **bass** solo after he ...
 - ... lovely **bass** voice
 - reach the **bass** notes ...
 - played **bass** guitar and ...
 - with a **bass** and a lead guitarist ...

Decompositional Approaches

Lexical Decomposition

Theory of lexical decomposition assumes that lexical meaning is complex. This complexity is expressed in structured meaning representations:

Jill likes Jack most of the time

What did Jill do for Jack to love him

- Predicates (likes)
- Arguments (linguistic) (Jill & Jack)
- Operators (linguistic) (what)

Decompositional Representations

One traditional approach (particularly associated with Katz and Fodor): definitions provide **necessary** and **sufficient conditions**

- Ex. One is a “bachelor” iff {human, male, adult, never married}
- A challenge: it’s *very* hard to identify necessary and sufficient conditions for any concept.
 - Is having four legs a necessary condition for something to be a dog? Does that mean a dog that’s lost one of its legs (say, in an accident) no longer a dog?

Decompositional Representations

Structured conceptual representations:

Jackendoff (2002): “Conceptual structure is not a part of language per se – it is a part of thought. It is the locus for the understanding of linguistic utterances in context, incorporating pragmatic considerations and “world knowledge”; it is cognitive structure in terms of which reasoning and planning take place”.

Decompositional Representations

Structured conceptual representations:

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Decompositional Representations Conceptual structure elements: categories, functions, arguments

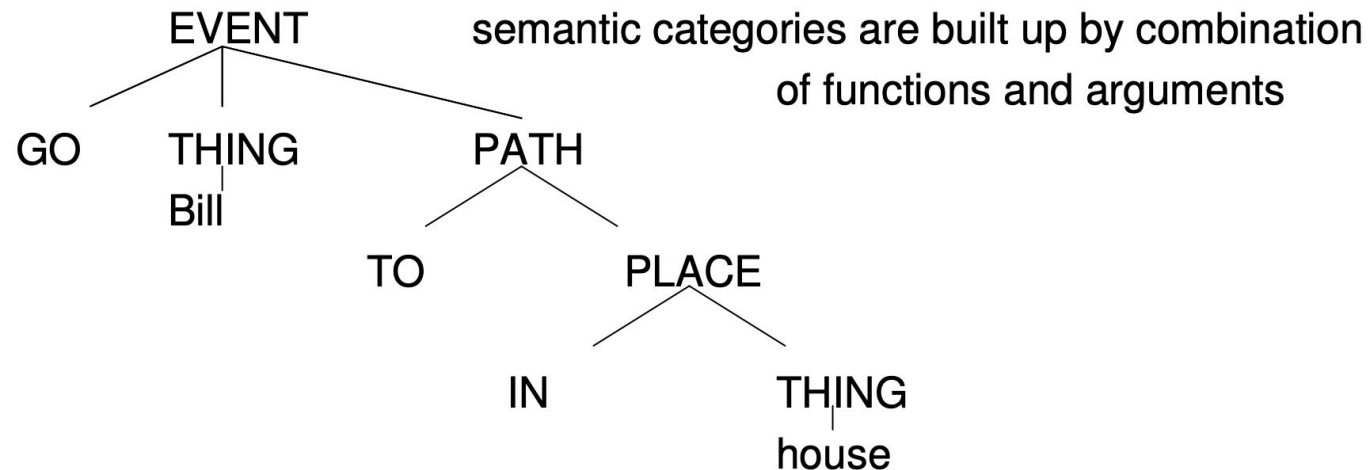
- Universal semantic categories: Event, State, Object (Thing), Path, Place, Property.

a. Syntax and phonology:

[John]_i went_j [into_k [the room]_m]

b. Conceptual structure:

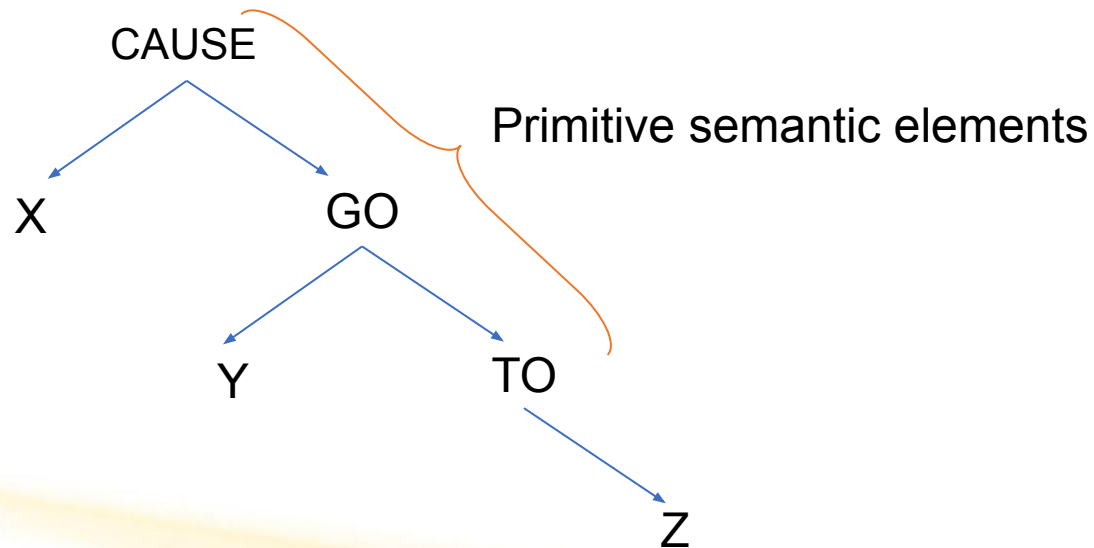
[_{Event} GO ([_{Thing} JOHN]_i,]_{Path} TO ([_{Place} IN ([_{Thing} ROOM]_m))]_k)]_j



Decompositional Representations

- Another traditional approach: structured conceptual representations
- A good example: lexical-conceptual structure (LCS)
 - Framework for semantic analysis, developed by Ray Jackendoff

give:



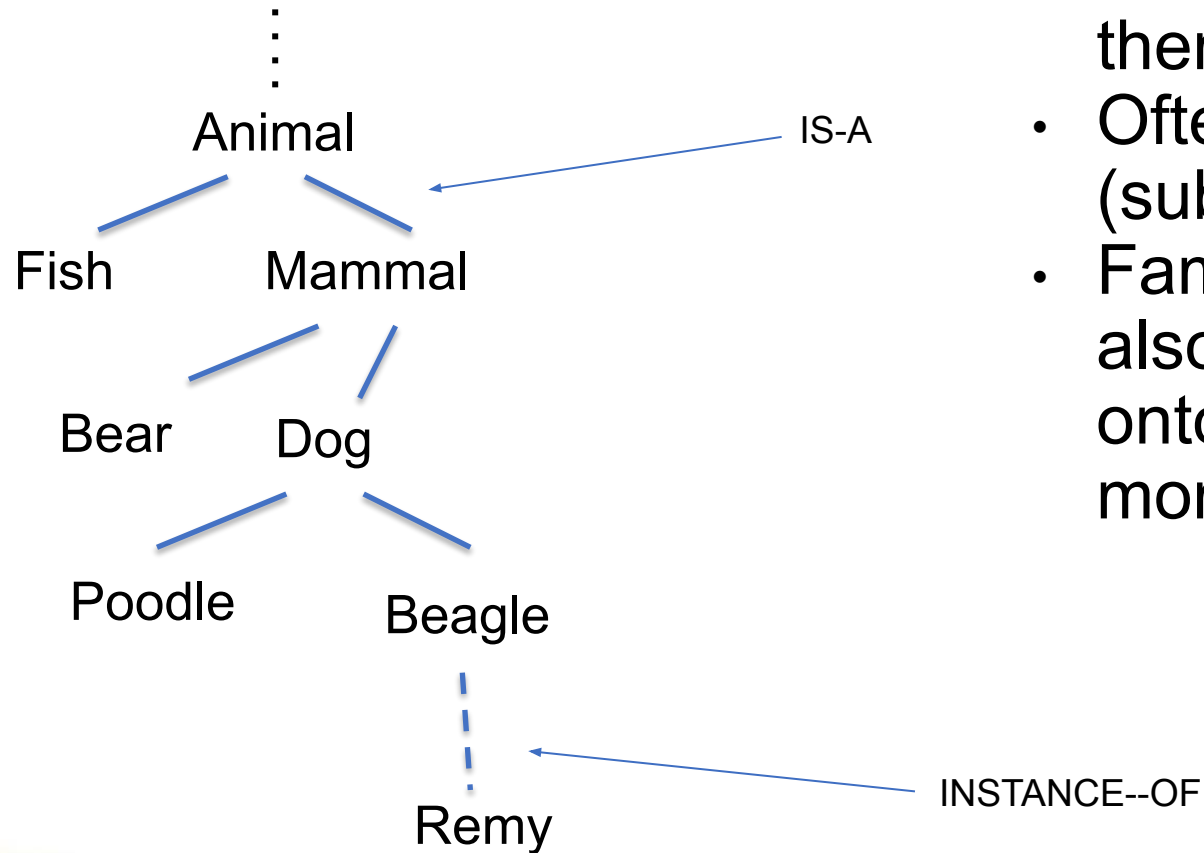
give: [CAUSE (x, [GO (y, [TO (z)]))]]

Ex: John *gave* me the book

John (=X) CAUSED the book (=Y) to go to me (=Z)

Ontological Approaches

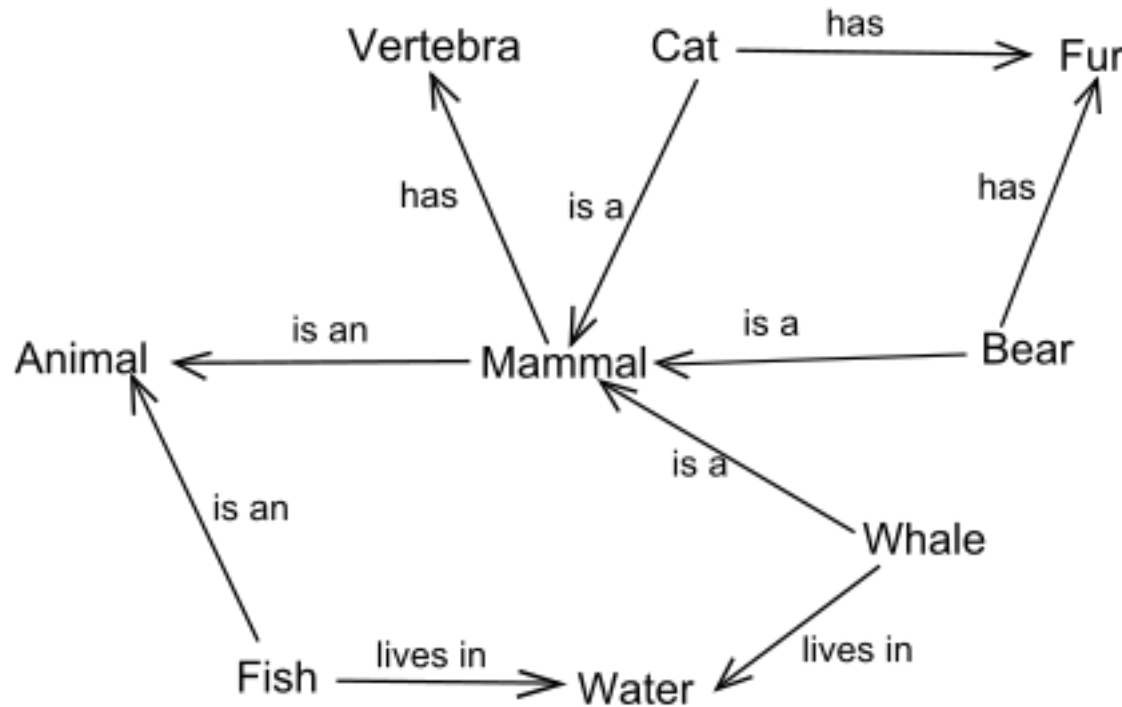
Ontologies



- Sets of entities and relations between them
- Often structured around the **IS-A** (subsumption) relationship for concepts
- Familiar from biology – but there are also gene ontologies, business ontologies, astronomy ontologies, and more.

Semantic Networks

- Also describes a sets of entities and relations between them
- Sometimes used interchangeably with “ontology”; the term “semantic network” tends to be used with knowledge structures where there is additional emphasis on slot/filler relationships (e.g. *cat HAS fur*)



Public Domain, <https://commons.wikimedia.org/w/index.php?curid=1353062>

WordNet

- Conceived by George Miller as an ontology for English
- Has evolved into a broader collection of multilingual WordNets
- Distinct ontologies organized by part of speech: N, V, Adj, Adv
- Core concept: synonym set (synset) ~ “Concept”
 - <board, plank, beam>
 - <board, committee, panel, council>
- The noun taxonomy is most widely used
 - Hyper/hyponym (IS-A)
 - Instance (INSTANCE-OF)
 - Meronym (PART-OF)
 - Antonym (OPPOSITE)
 - ...

George A. Miller, who began the WordNet project in the mid-1980s, passed away on July 22, 2012 at the age of 92.

WordNet

- IS-A is related to the core idea in AI of “inheritance”, also a fundamental role in early semantic networks
 - $C2 \text{ IS-A } C1 \Rightarrow \text{for all } f, f \text{ is a property of } C1 \Rightarrow f \text{ is also a property of } C2$
- Related to subclassing in object-oriented programming

Word Sense Disambiguation (WSD)

WSD

- A classic problem in NLP
- Bar-Hillel (1960) argued that fully-automatic high-quality machine translation (FAHQMT) was infeasible because it would require too much world knowledge; illustrated with word sense ambiguity
 - “The box is in the pen” requires commonsense reasoning about relative sizes to disambiguate which sense of “pen” is intended and translate
- Long era of work on small sets of individual words (e.g. line, bank)
- Resnik and Yarowsky (1997) instigated SENSEVAL (later SemEval)
 - Community-wide shared task for WSD
 - Evolved into broader range of semantic evaluation tasks

WSD as supervised classification

- Given:
 - enumerated senses $\{s_1, s_2, \dots, s_n\}$ for a word w
 - context for w (.... w )
- Select:
 - “correct” s_i for w
- Strong baseline: Just the most frequent sense!
- Good pre-deep-learning baseline: Naive Bayes
 - Choose $\operatorname{argmax}_y \Pr(y|x)$
(Where y ranges over senses. x is the word.)

WSD as supervised classification

- Strong baseline: Just the most frequent sense!
- Good pre-deep-learning baseline: Naive Bayes
 - Choose

$$\operatorname{argmax}_y \Pr(y \mid x) = \operatorname{argmax}_y \frac{\Pr(x \mid y) * \Pr(y)}{\Pr(x)}$$

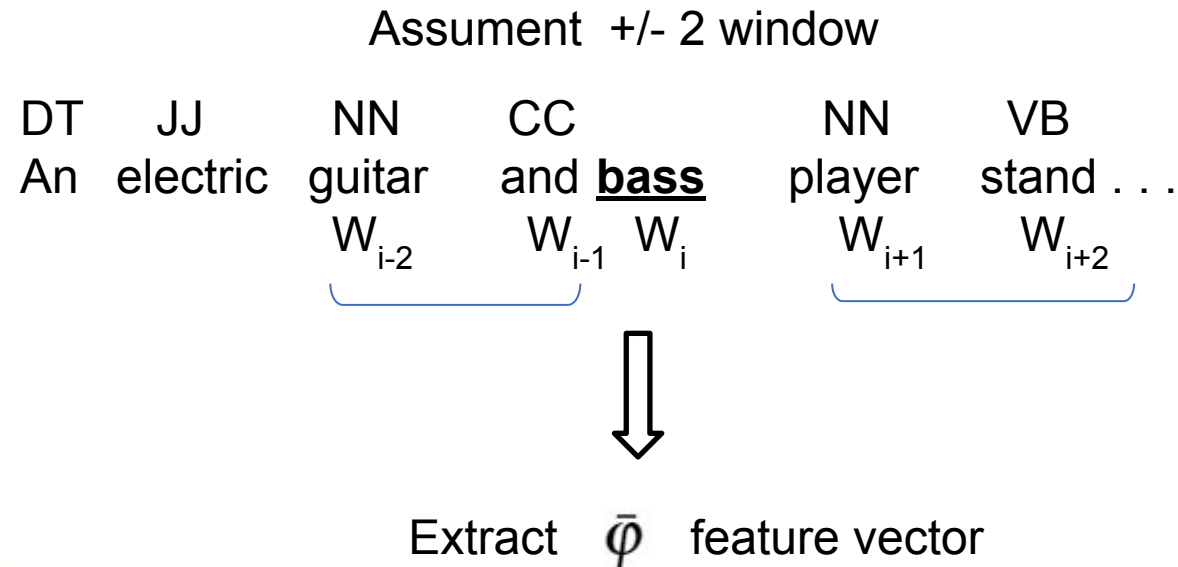
- Naively assume all features are independent

$$\operatorname{argmax}_y \underbrace{\prod_j (\Pr \varphi_j(x) \mid y)}_{\text{Multiply independent probabilities}} \cdot \underbrace{\Pr(y)}_{\text{Prior (often uniform)}}$$

(Where y ranges over senses. x is the word, and $\bar{\varphi}$ includes features of the context)

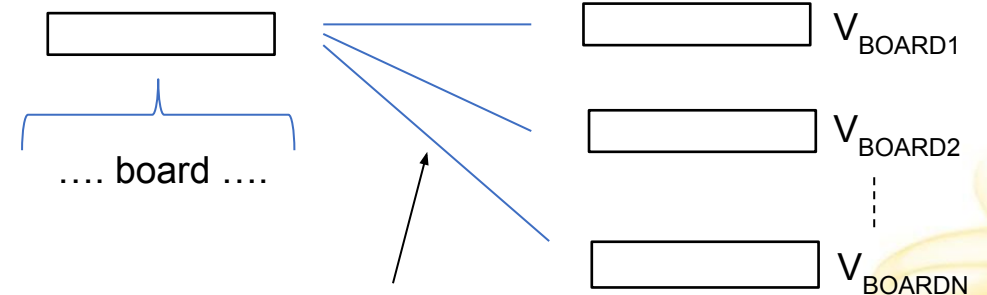
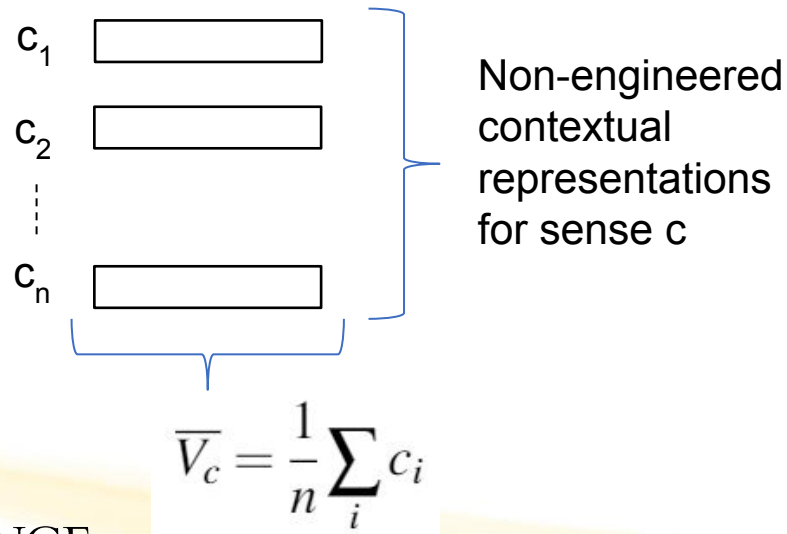
WSD as supervised classification

- Other supervised approaches:
 - SVM with engineered features



Supervised Method for WSD

- Given
 - ... sawed a **board** in half and ...
 - ... nailed together two pine **boards** ...
- Create contextual embedding vectors for each sense, and use 1-nearest-neighbor classifier!



Minimize cosine distance to select BOARD_i

“I don’t believe in Word Senses”

Kilgariff (2003)

Abandoning enumerated senses

- Kilgariff advocated for task-dependent clustering of corpus instances
- Schütze: Early distributional representations can form clusters
- Current trend is to use distributional representations as meaning
 - “You shall know a word by the company it keeps”
 - However, consider: systematic regularities (e.g. lamb is a food or animal?), sparse data (esp for specialized domains), explainability (supporting understanding and trusting inferences)

Why doesn't traditional WSD help a lot of standard applications?

- Low performance - hard problem...
- Skew of senses
 - Most-frequent sense is largely represented by the word itself (Zipf)
 - “Pen” usually means writing implement, for example
 - One sense per discourse
 - Implicit disambiguation via co-occurrence with other words
 - E.g. in IR: *bank* -> ambiguous, but *bank atm hours* -> unambiguous)

Distributional Approaches to Meaning

Distributional approaches to meaning

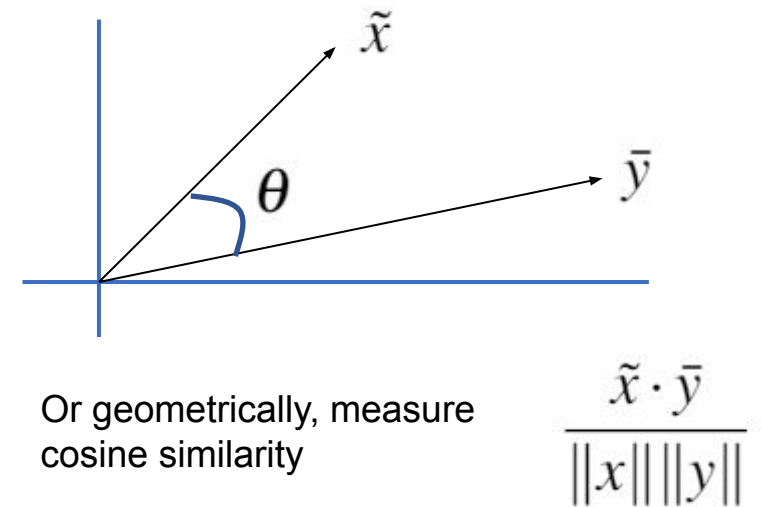
- Symbolic representations are basically 1-hot encodings

$$\bar{x}_{\text{apple}} \begin{bmatrix} 0 & \dots & 1 & \dots & 0 & \dots & 0 \\ \text{a} & \text{an} & \text{apple} & \dots & \text{orange} & \dots & \text{zoo} \end{bmatrix}$$

$$\bar{x}_{\text{orange}} \begin{bmatrix} 0 & \dots & 0 & \dots & 1 & \dots & 0 \end{bmatrix}$$

$$\bar{x}_{\text{zoo}} \begin{bmatrix} 0 & \dots & 0 & \dots & 1 & \dots & 0 \end{bmatrix}$$

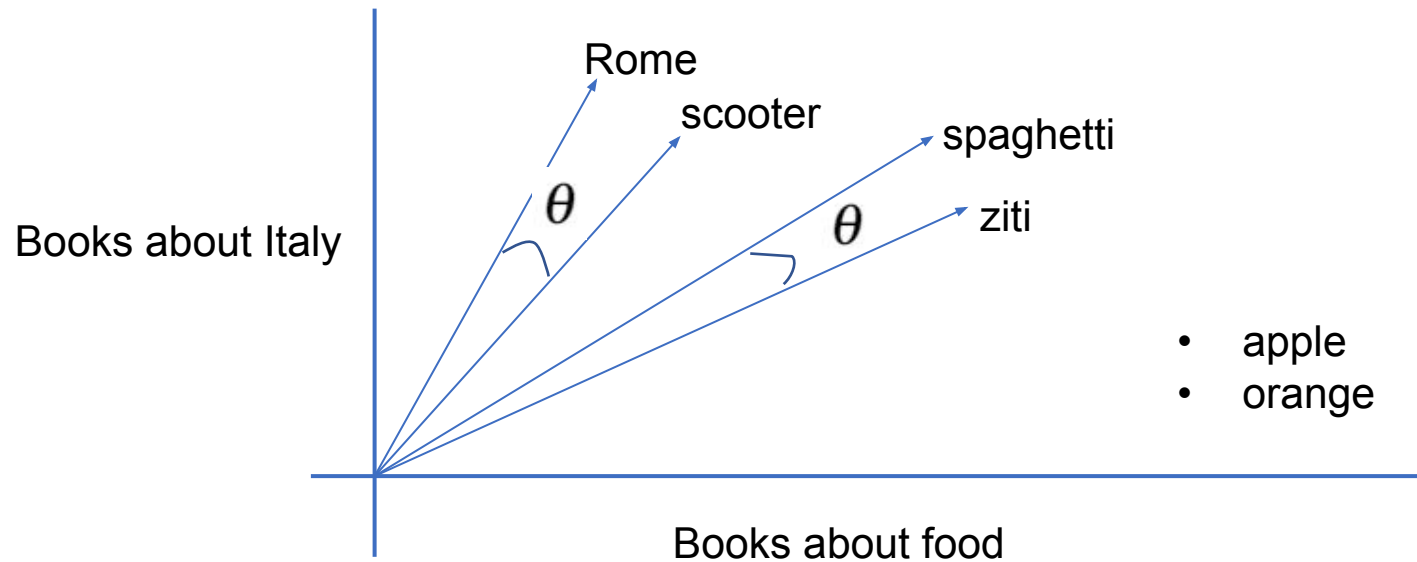
$$\bar{x} \cdot \bar{y} = \sum_{i=1}^n x_i y_i$$



Multiply 1xn matrix by nx1 matrix to get a 1x1 result

Vector Similarity

- Often measured as $\cos(\hat{x}, \hat{y}) = \hat{x} \cdot \hat{y}$



Problem: $\text{sim}(x,y) = 0$ for $x \neq y$ if 1-hot! Has trouble generalizing {puppy, dog, pup}

<https://machinelearningmastery.com/one-hot-encoding-for-categorical-data/>

Vector Similarity

- Problem: $\text{sim}(x,y) = 0$ for $x \neq y$ if 1-hot! Has trouble generalizing {puppy, dog, pup}
- Solution: Use weights (term frequency–inverse document frequency)

$$V_{w_i d_j} = + f_{i,j} \cdot \text{idf}_i$$

Freq. of w_i in d_j

inversely proportional to
of d_j 's containing w_i

Usually:

$$\log(\text{count}(w_i, d_j) + 1)$$

Usually: $\log(N / \# \text{ docs containing } w_i)$

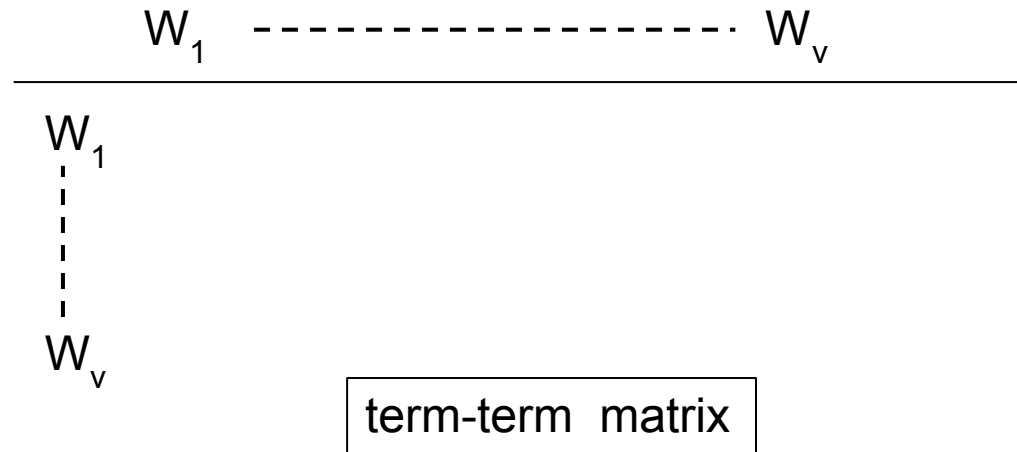
N: Number of documents

Ex. If “the” is in 99% of docs, its idf is really low

If $w_i \in$ all docs, $\text{idf} = 0$! Thus, denominator usually is $1 + \# \text{ docs containing } w_i$

Vector Similarity

- Using contexts to define word vectors



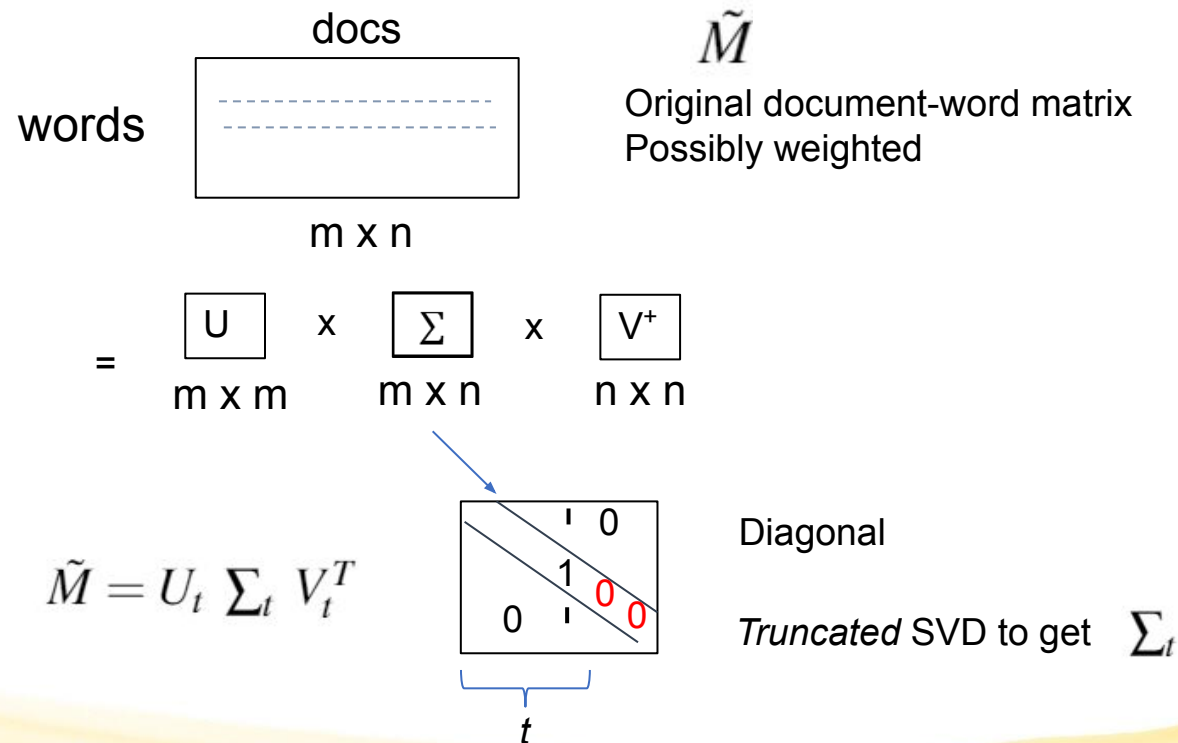
- $\text{count}(w_i, w_j) = \# \text{ times } w_i, w_j \text{ co-occur in a window}$
- Same issue with frequent words, PMI

$$I(x, y) = \log_a \frac{P(x, y)}{P(x) \cdot P(y)}$$

$$PPMI = \max(PMI, 0)$$

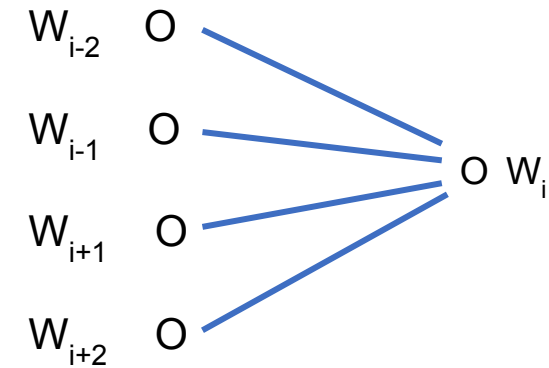
Dimensionality Reduction

- Each word has a very high dimension! This can lead to reduced ability to generalize.
- What are some dimension reduction techniques we can leverage?
 - Latent Semantic Indexing (LSI): use Singular Value Decomposition to reduce words/documents.



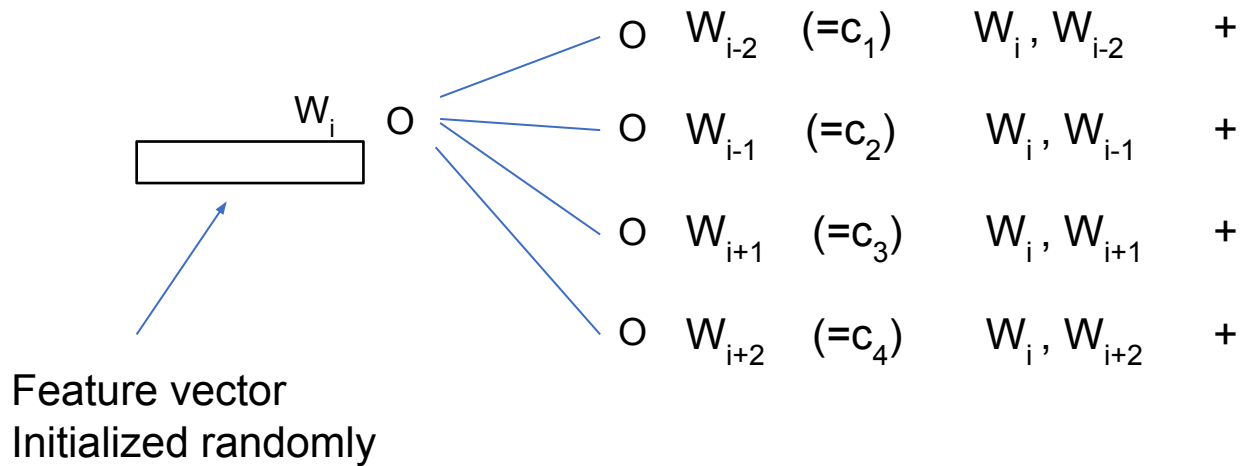
word2vec

- CBOW - simple! Predict target word from context



- Skip-gram - a bit more involved, predict context words from target (reverse CBOW)

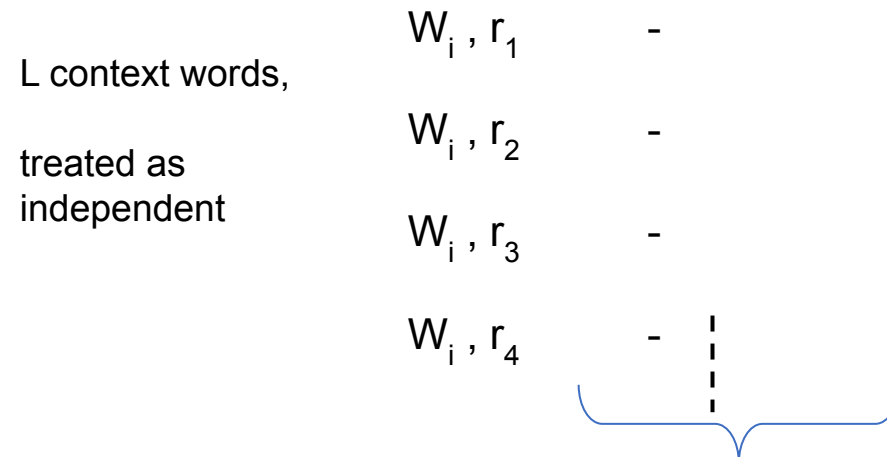
word2vec: skip-grams



- Goal: predict context words for a given target word
- Optimize feature vectors:
 - Intuition: we want context words with target to be high, and random words with target to be low

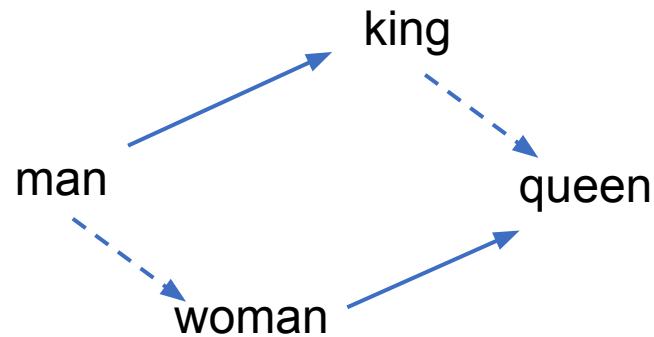
$$\sigma(c_j \cdot w) \Rightarrow \Pr(+ \mid c_j, w)$$

$$\sigma(r_m \cdot w) \Rightarrow \Pr(- \mid r_m, w)$$



Effect of word2vec

- Words that appear in similar contexts get closer-together feature vectors
- Smaller contexts: more syntactic; context words more likely to be syntactically related
- Wider contexts: tend to capture more topical aspects, “aboutness”
- Captures structure in semantic spaces (classic man -> king as woman -> queen example)



- Can extend to document-level work (e.g. doc2vec!)